Activity 6.2: Circumference, Area & Equations of Circles

A

Name: _____ Partner's name: _____ Date: _____

Directions: Give exact answers (in terms of π) when needed. Diagrams are accurate, but use math rather than a ruler or protractor.

Work alone, then compare after each problem. If answers differ, find the mistake together.

1 Find the circumference of a circle with radius 7.

Answer: _____ Match? ☐ Yes ☐ No

2 Find the area of a circle with radius 5.

Answer: _____ Match? ☐ Yes ☐ No

3 A circle has area 49π. Find its radius.

Answer: _____ Match? ☐ Yes ☐ No

4 Write the equation of the circle with center $(0, 0)$ and radius 5.

Answer: _____ Match? ☐ Yes ☐ No

5 Find the center and radius: $(x-3)^2+(y+2)^2 = 16$

Answer: _____ Match? ☐ Yes ☐ No

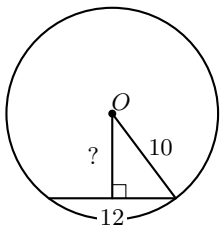
6 Write the equation of the circle with center $(-1, 4)$ and radius 3.

Answer: _____ Match? ☐ Yes ☐ No

7 The point $(3, y)$ is in Quadrant I and lies on the circle $x^2 + y^2 = 25$. Find y.

Answer: _____ Match? ☐ Yes ☐ No

8 Find the distance from the center to the chord.



Answer: _____ Match? ☐ Yes ☐ No

Activity 6.2: Circumference, Area & Equations of Circles

B

Name: _____ Partner's name: _____ Date: _____

Directions: Give exact answers (in terms of π) when needed. Diagrams are accurate, but use math rather than a ruler or protractor.

Work alone, then compare after each problem. If answers differ, find the mistake together.

1 Find the circumference of a circle with diameter 14.

Answer: _____ Match? ☐ Yes ☐ No

2 Find the area of a circle with diameter 10.

Answer: _____ Match? ☐ Yes ☐ No

3 A circle has circumference 14π. Find its radius.

Answer: _____ Match? ☐ Yes ☐ No

4 Write the equation of the circle with center $(0, 0)$ that passes through $(3, 4)$.

Answer: _____ Match? ☐ Yes ☐ No

5 Find the center and radius: $x^2-6x+y^2+4y-3=0$

Answer: _____ Match? ☐ Yes ☐ No

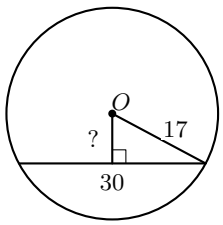
6 Write the equation of the circle with center $(-1, 4)$ that passes through $(2, 4)$.

Answer: _____ Match? ☐ Yes ☐ No

7 The point $(2, y)$ is in Quadrant I and lies on the circle $x^2 + y^2 = 20$. Find y.

Answer: _____ Match? ☐ Yes ☐ No

8 Find the distance from the center to the chord.



Answer: _____ Match? ☐ Yes ☐ No